

VIRTUAL PRIVATE CLOUD (VPC) -III

(NAT Gate way)

❖ NAT Gateway

- A NAT Gateway is a managed service in **AWS** that **allows** EC2 instances in private subnets to **access the internet** or other AWS services, **without allowing** anyone from the internet to access them.
- ✓ Key Characteristics

1. Purpose

- Helps **private instances** connect to the internet.
- Keeps private instances **secure and hidden** from outsiders.
- Allows only **outbound (outgoing) internet traffic**, not inbound.
- Works with both **IPv4 and IPv6**

2. Architecture Components

- **Public Subnet Placement:** NAT Gateway must be created inside a public subnet (itself needs internet access).
- **Elastic IP:** It must be assigned at least one Elastic IP (a fixed public IP).
- **Availability Zone (AZ) Specific:** NAT Gateway belongs to one AZ. It is not multi-AZ by default.
- **Route Tables:** You must update route tables in private subnets to send traffic to NAT Gateway.

3. How Traffic Flows through NAT Gateway

1. Private instance wants to **access the internet** (like to update software).
2. It **sends the request** to the NAT Gateway.
3. NAT Gateway **changes the instance's** private IP to a public IP (using Elastic IP).
4. Request **goes** to the internet.
5. Internet **sends back the response**.
6. NAT Gateway **changes the public IP** back to the private IP.
7. The response **reaches the private instance**.

❖ Comparison: NAT Gateway vs NAT Instance

Feature	NAT Gateway	NAT Instance
Managed by AWS	✓ Yes (fully managed)	✗ No (user-managed EC2 instance)
Setup Complexity	✓ Easy (create + attach to subnet)	✗ Manual (launch EC2, configure routes, etc.)
Scalability	✓ Automatically scales	✗ Manual scaling needed
Availability	✓ Highly available (with multiple AZs)	✗ You must configure HA with scripts/load balancers
Performance	✓ High bandwidth (up to 45 Gbps)	✗ Limited by EC2 instance size/type
Security Groups	✗ Not supported	✓ Can attach SG to control traffic
Cost	💰 More expensive	💰 Cheaper (for low traffic or dev environments)

1. Troubleshooting Tips

1. Common Problems

- Route table not pointing to NAT Gateway
- Elastic IP not assigned or released by mistake
- Hitting bandwidth or connection limits
- Unexpected high cost due to forgotten or misused NAT Gateway

2. How to Fix

- Check route table entries in private subnet
- Confirm Elastic IP is properly associated
- Use CloudWatch logs to monitor performance and traffic
- Review billing reports for data transfer costs

❖ AWS VPC Peering

VPC Peering is a **network connection between two Virtual Private Clouds (VPCs)** that allows them to **communicate privately using private IP addresses**, as if they were within the same network.

▪ **Key Characteristics**

1. Purpose

- Allows communication between **two VPCs** using private IPs.
- Helps you share services like databases, APIs, internal tools.
- Works for both **IPv4 and IPv6** traffic.
- Maintains **privacy and security** (no internet exposure)

▪ **Network Flow (How data moves)**

1. Instance in **VPC A** sends a request to an instance in **VPC B**.
2. The request is routed based on **VPC A's** route table.
3. Traffic flows through the **VPC peering** connection.
4. **VPC B's** route table sends the request to the target instance.
5. The response follows the **same path** back

Use Cases

1. Common Scenarios

- Multi-tier applications where **backend and frontend** live in different VPCs.
- A shared services **VPC** (like one common database or authentication service).
- Connecting development, testing, and production environments.
- **Allowing access** between VPCs owned by different AWS accounts.

2. When to Use

- When **two VPCs need** to communicate internally using private IPs.
- In microservices setup, where services are split into **different VPCs**.
- When you want secure **communication between** environments.
- For **cross-account** resource sharing between teams or departments.

Cross-Account VPC Peering

Requirements

- CIDR blocks must be **non-overlapping**.
- Both accounts must have correct **IAM permissions**.
- Route tables and security groups must be configured correctly in both VPCs.
- Both sides (requester and acceptor) must agree **and approve the peering**.

Process

- **Requester** sends a peering request.
- **Acceptor** approves it from their AWS account.
- Route tables are updated in both VPCs.
- Security groups are configured to allow communication.
- Test connectivity between instances

[GENERAL INTERVIEW QUESTION]

Q1. How do you make a subnet public in a VPC?

- *Attach an Internet Gateway to the VPC*
- Create or update the subnet's route table with a route: 0.0.0.0/0 → igw-xxxx
- Enable auto-assign public IP on EC2 launch

Q2. Can a VPC have multiple IGWs? Why or why not?

- No, each VPC can only have one IGW. This ensures simpler routing and avoids asymmetric or misrouted traffic issues

Q3. What is the difference between local and IGW route entries in a VPC route table?

- **Local** route allows intra-VPC communication
- **IGW** (0.0.0.0/0) route enables internet access

Q4. What is the difference between a public and private EC2 instance in AWS?

- **Public EC2:** Has a public IP and subnet routes to IGW
- **Private EC2:** No public IP and no IGW route—used for internal apps, DBs, etc.

Q5. How do you SSH into a private instance securely?

- Via a **Bastion Host** in the public subnet, or using AWS Systems Manager (SSM) **Session Manager** for secure, keyless access.

Q6. Can a public instance communicate with a private instance in the same VPC?

- Yes, as long as **Security Groups** and **NACLs** allow the traffic.

Q7. How do you ensure public EC2 instances are secure?

- **Restrict** Security Group rules
- Disable root login
- Use key pair authentication
- Restrict SSH to specific IPs (e.g., office IP)

Q8. What is the purpose of a NAT Gateway in AWS?

- To **allow private EC2** instances to access the internet outbound only (e.g., for OS updates) without receiving inbound traffic.

Q9. What is VPC Peering in AWS?

- A **private connection between two VPCs** that allows traffic to route between them using private IP addresses.

Q10. What are the limitations of VPC Peering?

- No transitive routing
- No overlapping CIDRs
- Manual route and security configuration
- Doesn't support security group referencing across VPCs